

Aim:

Write a Python program that reads a series of integers entered by the user, stores them in a tuple, and performs the following operations:

- Print the total number of items in the tuple.
- Print the last item of the tuple.
- Print the tuple elements in reverse order.
- Check whether the integer 5 is present in the tuple and print **Yes** if found; otherwise, print **No**.
- Remove the first and last items from the tuple, sort the remaining elements, and print the resulting tuple.

Input Format:

- The input consists of a single line containing space-separated integers.
- The program displays the following prompt:

Enter integers separated by space:

Output Format:

Display the following information:

```
Total number of items: <count>
Last item: <last_item>
Tuple in reverse order: <reversed_tuple>
Yes/No
Sorted tuple after removing first and last items: <sorted_tuple>
```

where:

- <count> is the total number of elements in the tuple.
- <last_item> is the last element of the tuple.
- <reversed_tuple> is the tuple printed in reverse order.
- **Yes** is displayed if the tuple contains 5; otherwise, **No** is displayed.
- <sorted_tuple> is the sorted tuple obtained after removing the first and last elements.

Sample Input:

Enter integers separated by space: 1 2 3 4 5

Sample Output:

```
Total number of items: 5
Last item: 5
Tuple in reverse order: (5, 4, 3, 2, 1)
Yes
Sorted tuple after removing first and last items: (2, 3, 4)
```

Source Code:

input_tuple_analysis.py

```
#write your code here...
n=tuple(map(int,input("Enter integers separated by space: ").split()))
```

```

print("Total number of items:",len(n))
print("Last item:",n[-1])
print("Tuple in reverse order:",n[::-1])
if 5 in n:
    print("Yes")
else:
    print("No")
print("Sorted tuple after removing first and last items:",tuple(sorted(n[1:-1])))

```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter integers separated by space:
10 5 8 3 12 7
Total number of items: 6
Last item: 7
Tuple in reverse order: (7, 12, 3, 8, 5, 10)
Yes
Sorted tuple after removing first and last items: (3, 5, 8, 12)

Test Case - 2
User Output
Enter integers separated by space:
1 2 3 4 6
Total number of items: 5
Last item: 6
Tuple in reverse order: (6, 4, 3, 2, 1)
No
Sorted tuple after removing first and last items: (2, 3, 4)

Test Case - 3
User Output
Enter integers separated by space:
5 9 2 8 1
Total number of items: 5
Last item: 1
Tuple in reverse order: (1, 8, 2, 9, 5)
Yes
Sorted tuple after removing first and last items: (2, 8, 9)

Test Case - 4
User Output
Enter integers separated by space:
4 7 5 6 2 9
Total number of items: 6
Last item: 9
Tuple in reverse order: (9, 2, 6, 5, 7, 4)
Yes
Sorted tuple after removing first and last items: (2, 5, 6, 7)

Test Case - 5
User Output
Enter integers separated by space:
11 22 33 44 55
Total number of items: 5
Last item: 55
Tuple in reverse order: (55, 44, 33, 22, 11)
No
Sorted tuple after removing first and last items: (22, 33, 44)